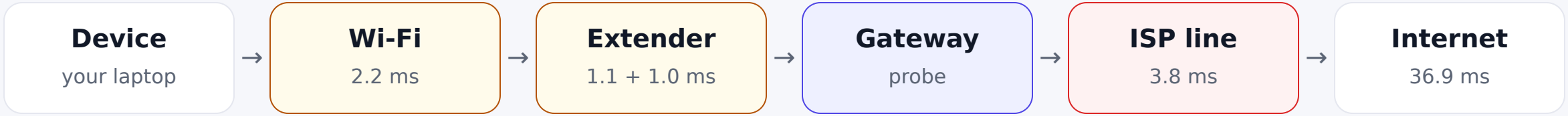
Heimdall latency health check for ISPs and their customers

When the internet feels slow, who should fix it?

A small probe on the home gateway and the Wi-Fi extender measures every part of the path, and tells within seconds where the delay is and who should act.



Probes on the gateway and the extender · typical latency of each part on 30 Oct, from the daily analysis

1 · Measure

Every client connection's Wi-Fi and internet round trips, the extender link, the ISP's hops, 37 services, and the same services from AWS for comparison.

2 · Detect

Baselines per device and service. A slowdown becomes an incident with a likely cause: weak Wi-Fi, the extender, the ISP's line, its route to a service, DNS, or the service itself.

3 · Tell

A Slack alert, and a status page for the ISP and for the customer: what happened, why, who should act and the fix. Grafana for us, a weekly report for the home.

No hostnames or server addresses are stored · 8 MB static binary on a Zyxel EX7501 gateway and a WX5600 extender · AWS: Lambda, Firehose, Athena, DynamoDB, Grafana

Heimdall live demo

Slow down my laptop, and watch it get caught

1 Slow it down `demo slow 40`

Adds 40 ms to my laptop's traffic. It is on the extender's Wi-Fi.

2 Caught in ~10 s `Slack`

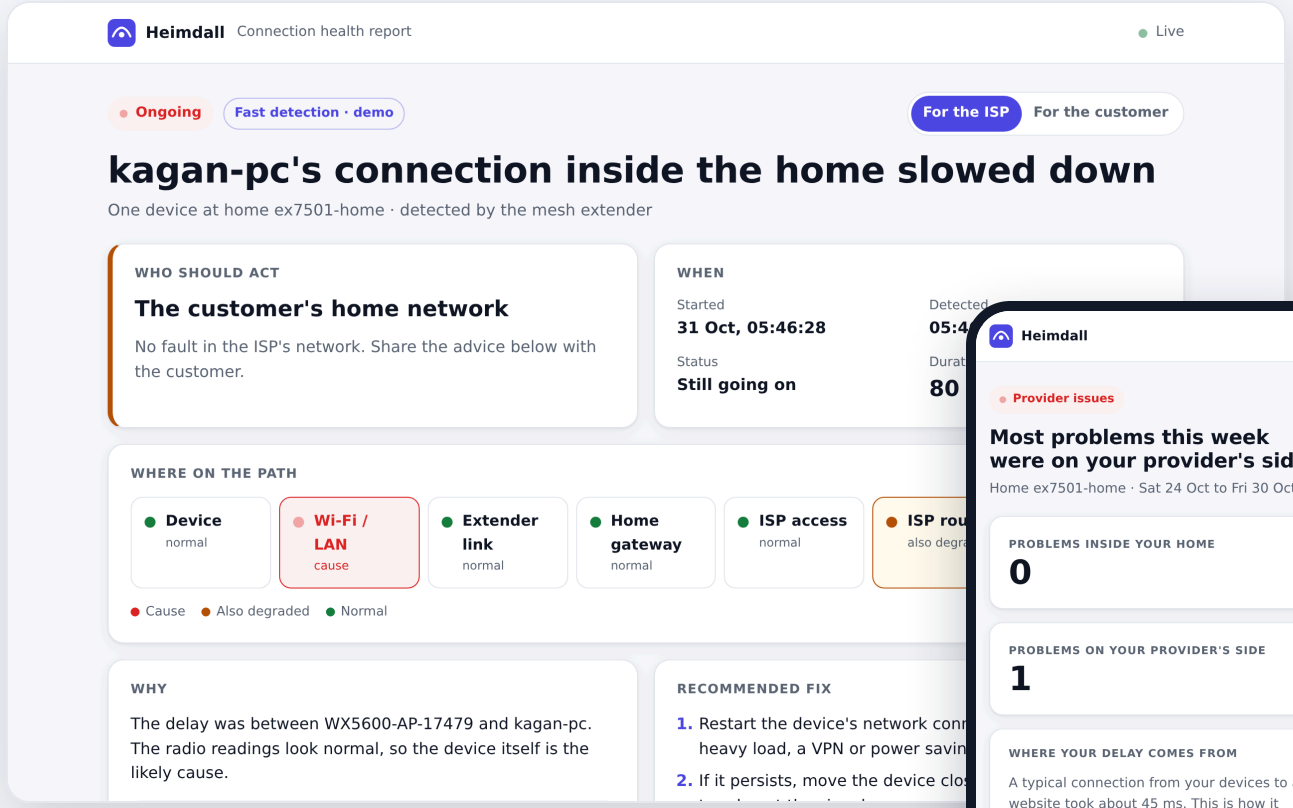
The extender's probe sees kagan-pc's hop jump from 1 ms to 41 ms.

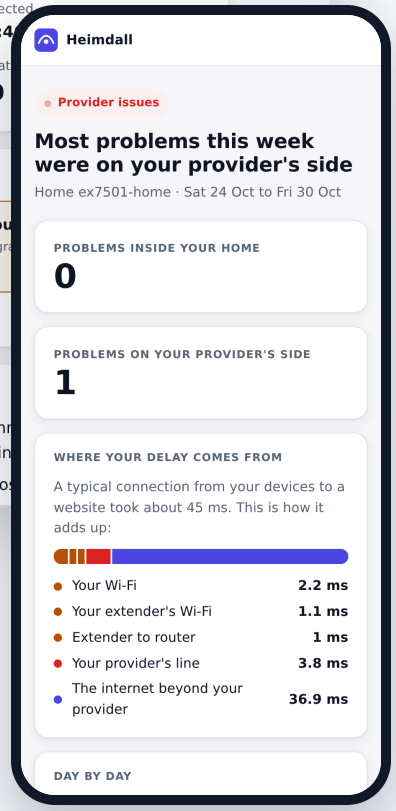
3 Who should act `status page`

The customer's home network, not the ISP: the ISP's line, its route and the services were normal.

4 Fix it `demo fix`

Resolved about 20 s later, with the real duration.





For the ISP: the cause, who should act, the fix
For the customer: the same in their words, and a weekly report of their home

Demo mode runs a second detector on 5-second buckets. The regular one waits 10 minutes before it alerts.